

All documents, calculators and mobile phones prohibited.

Particular consideration will be given to the care taken in writing.

### Exercise 1 : 10 pts

- 1 Using the truth table show that the logical implication is a transitive connective, that is :  
 $((P \Rightarrow Q) \wedge (Q \Rightarrow R)) \Rightarrow (P \Rightarrow R)$ .
- 2 For any natural number  $n$  :
  - a Expand the expression :  $(n-1)n(n+1)(n^2+1)$ .
  - b Prove that : 5 divides  $(n^5 - n)$ .
- 3 Let  $S$  be a finite set with cardinality  $|S| = n \in \mathbb{N}$ .  
 Prove by induction that  $|\mathcal{P}(S)| = 2^n$  ( $\mathcal{P}$  denotes the power set).
- 4 We define on  $\mathbb{R}^2$  the binary relation  $\mathfrak{R}$  by :  $((x, y) \mathfrak{R} (x', y')) \Leftrightarrow (\exists \alpha \in \mathbb{R}^* : (\alpha x, \alpha^2 y) = (x', y'))$ .
  - a Show that  $\mathfrak{R}$  is an equivalence relation.
  - b Given  $(a, b) \in \mathbb{R}^2$ , define  $\overline{(a, b)}$  the class of equivalence of  $(a, b)$ ?
  - c Determine and represent graphically  $\overline{(1, 1)}$ .

### Exercise 2 : 10 pts

$e \simeq 2,718281828\dots$  is the base of the natural logarithm

- 1 Consider the function  $f : \mathbb{R} \rightarrow ]-1, 1[$  defined by  $f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ .
  - a Show that :  $\forall x \in \mathbb{R} : f(x) = 1 - \frac{2}{e^{2x} + 1}$ .
  - b Is  $f$  injective, surjective? justify using definitions.
- 2 Let  $G = ]-1, 1[$  and  $\varphi : [-1, 1] \rightarrow \mathbb{R}$  the strictly increasing function defined by
 
$$\varphi(x) = \frac{x+a}{ax+1} \text{ where } a \in G.$$
  - a Prove that  $G$  is closed under the binary operation  $\star$  defined by  $a \star b = \frac{a+b}{ab+1}$ .
  - b Establish that  $(G, \star)$  is an abelian group.
- 3 Show that  $f$  is a group isomorphism from  $(\mathbb{R}, +)$  to  $(G, \star)$ .

### Remark

- $f$  : is the **hyperbolic tangent** function  $\tanh$ .
- $\varphi$  : is a **Möbius** transformation.